Set 4: Product Inventory Management

```
library(ggplot2)
library(plotly)
library(shiny)
inventory <- data.frame(
  ProductID = c(1, 2, 3, 4, 5),
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  QuantityAvailable = c(250, 175, 300, 200, 220),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories")
)
print(inventory)
bar_chart <- ggplot(inventory,
            aes(x = ProductName,
              y = QuantityAvailable,
              fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity Available for Each Product",
     x = "Product Name",
     y = "Quantity Available") +
  theme_minimal()

print(bar_chart)
stacked_chart <- ggplot(inventory,
              aes(x = Category,
                y = QuantityAvailable,
                fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(title = "Quantity of Products by Category",
     x = "Category",
     y = "Quantity Available") +
  theme_minimal()
```

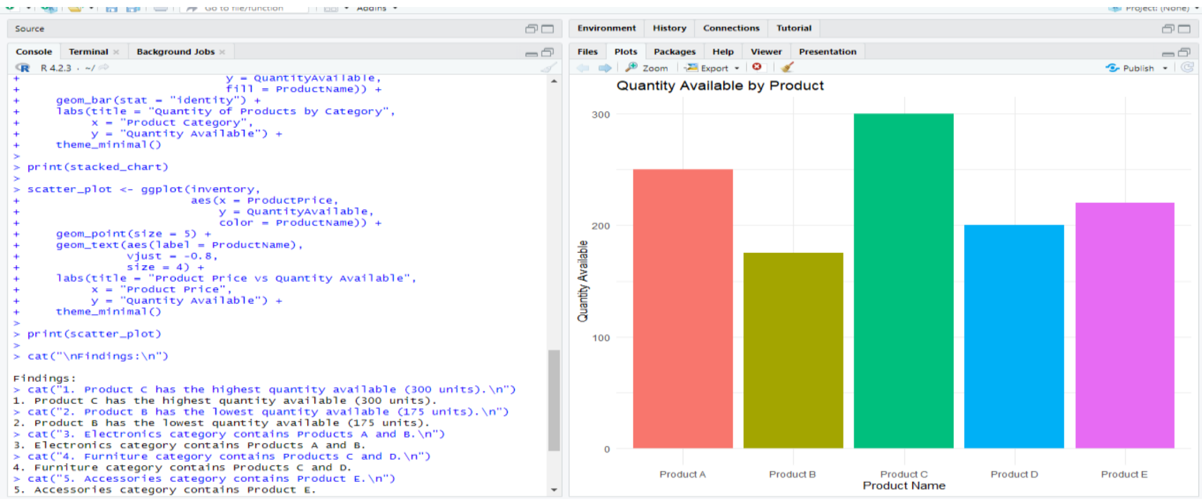
```
print(stacked_chart)
scatter_plot <- ggplot(inventory,
                aes(x = ProductID,
                  y = QuantityAvailable,
                  color = ProductName)) +
  geom_point(size = 4) +
  geom_line(group = 1) +
  labs(title = "Product ID vs Quantity Available",
     x = "Product ID",
     y = "Quantity Available") +
  theme_minimal()

print(scatter_plot)
ui <- fluidPage(

  titlePanel("Product Inventory Dashboard"),

  sidebarLayout(

    sidebarPanel(

      selectInput("category",
            "Select Category:",
            choices = c("All", unique(inventory$Category)),
            selected = "All")

    ),

    mainPanel(
```

```
          plotlyOutput("barPlot"),
          br(),
          plotlyOutput("stackPlot"),
          br(),
          plotlyOutput("scatterPlot")

        )
    )
)

server <- function(input, output) {

    filteredData <- reactive({

        if (input$category == "All") {
          inventory
        } else {
          subset(inventory, Category == input$category)
        }

    })

    output$barPlot <- renderPlotly({

        p <- ggplot(filteredData(),
                  aes(x = ProductName,
                      y = QuantityAvailable,
                      fill = ProductName)) +
          geom_bar(stat = "identity") +
          labs(title = "Bar Chart",
              x = "Product Name",
```

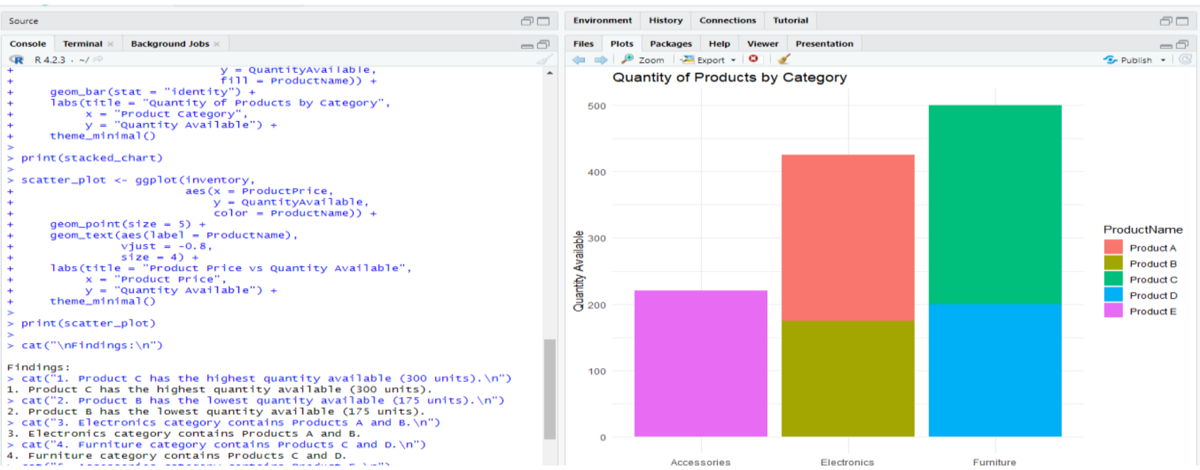
```
        y = "Quantity") +
      theme_minimal()

    ggplotly(p)

})

output$stackPlot <- renderPlotly({

    p <- ggplot(filteredData(),
              aes(x = Category,
                  y = QuantityAvailable,
                  fill = ProductName)) +
      geom_bar(stat = "identity") +
      labs(title = "Stacked Bar Chart",
          x = "Category",
          y = "Quantity") +
      theme_minimal()

    ggplotly(p)

})

output$scatterPlot <- renderPlotly({

    p <- ggplot(filteredData(),
              aes(x = ProductID,
                  y = QuantityAvailable,
                  color = ProductName,
                  text = ProductName)) +
      geom_point(size = 4) +
```

```
        geom_line(group = 1) +
        labs(title = "Scatter Plot",
             x = "Product ID",
             y = "Quantity Available") +
        theme_minimal()

      ggplotly(p, tooltip = c("text", "x", "y"))

  })

}

shinyApp(ui = ui, server = server)
```

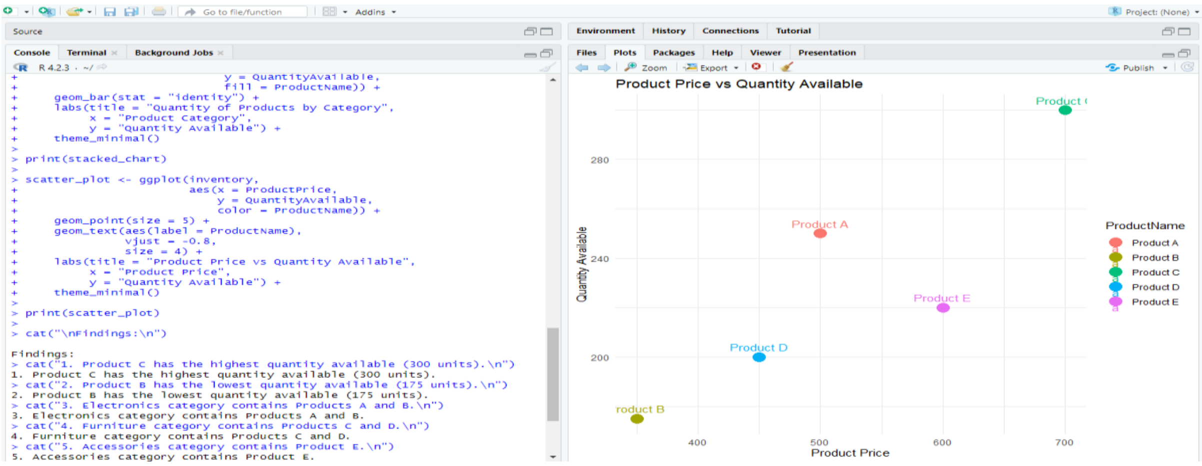
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



2. Generate a stacked bar chart to show the quantity of each product within different product categories.



3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.



4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.